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ORIGINAL ARTICLE

Machine-learning based prediction models for assessing skin irritation and corrosion potential of liquid chemicals using physicochemical properties by XGBoost

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Abstract
Skin irritation test is an essential part of the safety assessment of chemicals. Recently, computational models to predict the skin irritation draw attention as alternatives to animal testing. We developed prediction models on skin irritation/corrosion of liquid chemicals using machine learning algorithms, with 34 physicochemical descriptors calculated from the structure. The training and test dataset of 545 liquid chemicals with reliable in vivo skin hazard classifications based on UN Globally Harmonized System [category 1 (corrosive, Cat 1), 2 (irritant, Cat 2), 3 (mild irritant, Cat 3), and no category (nonirritant, NC)] were collected from public databases. After the curation of input data through removal and correlation analysis, every model was constructed to predict skin hazard classification for liquid chemicals with 22 physicochemical descriptors. Seven machine learning algorithms [Logistic regression, Naïve Bayes, k-nearest neighbor, Support vector machine, Random Forest, Extreme gradient boosting (XGB), and Neural net] were applied to ternary and binary classification of skin hazard. XGB model demonstrated the highest accuracy (0.73–0.81), sensitivity (0.71–0.92), and positive predictive value (0.65–0.81). The contribution of physicochemical descriptors to the classification was analyzed using Shapley Additive exPlanations plot to provide an insight into the skin irritation of chemicals.

Keywords Skin irritation/corrosion · Physicochemical descriptor · Machine learning · Extreme gradient boosting (XGB) · Shapley Additive exPlanations (SHAP)

Introduction

Assessment of skin irritation/corrosion potential is mandated for new cosmetic and chemicals [1]. According to the United Nations (UN) Globally Harmonized System of Classification and Labelling of Chemicals (GHS), skin hazard is largely classified into the skin corrosion, which refers to the production of irreversible skin damage with visible necrosis through the epidermis and into the dermis or skin irritation of reversible skin damage. The standard skin irritation test was the Draize rabbit skin test, OECD Test Guideline (TG) 404 [2]. However, due to the scientific, economic, and animal welfare reasons, alternative test methods are needed.

As an alternative to animal testing of skin irritation, the OECD TG 431 and 439 in vitro test methods using reconstructed human epidermis (RhE) test were approved for dermal acute toxicity [3, 4]. The UN GHS labels Category 1 (Cat 1) chemicals as skin corrosive, further subcategorized into Categories 1A, 1B and 1C depending on the in vivo effects during the exposure time and observation period. OECD TG 431 using RhEs classifies test chemicals into corrosives or non-corrosives [5]. On the other hand, OECD TG 439 is an in vitro test procedure to give the irritant (Cat 2) or non-irritant (no category) classification. OECD TG 439 uses same RhE test systems as TG 431, but protocol is different from TG 431. However, current OECD TGs do not allow classification of chemicals to the optional UN GHS Category 3 (Cat 3, mild irritants).

Importantly, a single in vitro test is insufficient to fully replace animal tests and combination of 2 or more in vitro

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tests are necessary to address full UN GHS category. Recently integrated approaches of in vitro and in silico methods are studied for reliable prediction of skin irritation/corrosion potential of chemicals [6–8]. Computational models such as read-across, structure–activity relationship (SAR), quantitative structure–activity relationship (QSAR) and machine-learning models have advantages of reducing cost and time for experimental test in the assessment of chemical toxicity [6, 9, 10]. QSAR modeling is a statistical, or machine learning approach based on linear or non-linear relationships between the toxicological activity and the structural parameters. Many (Q)SAR models are recently available for diverse toxic endpoints, but they need to be validated in accordance with the OECD's Guidance Document on the validation of (Q)SAR Models to fulfill regulatory purposes [11].

The performance of QSAR models, Derek Nexus, Toxtree and Case Ultra, was investigated for the prediction of skin and eye irritation/corrosion in the context of REACH [12]. Derek Nexus is based upon expert knowledge composed of alerts, examples, and reasoning rules to identify a specific chemical substructure inducing a specific toxicological effect [13]. Toxtree estimates the skin irritation and corrosion based on physicochemical descriptors (e.g., melting point, log P and lipid solubility) and structure rules [14]. Case Ultra is a fragment-based statistical model system, which breaks down the structures into all possible fragments and identifies the fragments considered as positive alerts for the toxicological effect. Unfortunately, the three models showed poor performances. Derek Nexus demonstrated a prediction for 109 compounds with an accuracy of 0.51, sensitivity of 0.14 and specificity of 0.96. Toxtree could not predict the skin irritancy of most of substances. Case Ultra predicted the skin irritation of 73 of 107 compounds with an accuracy of 0.59, sensitivity of 0.32 and specificity of 0.89, suggesting that more refined models shall be developed for the prediction of skin irritation/corrosion.

Here, we aimed to develop a machine learning model to predict the skin irritation/corrosion potential of liquid chemicals. Considering the lack of reliability of skin irritation/corrosion models, we have collected and preprocessed the in vivo database from ChemSkin DB [15] and eChemPortal to generate a high-quality curated dataset with reliable in vivo data. Physicochemical descriptors were calculated using ChemTunes:ToxGPS. After the curation of input data through removal and correlation analysis, every model was constructed to predict skin irritation classification for liquid chemicals (n = 545) with 22 physicochemical descriptors. Seven machine learning algorithms (logistic regression, Naïve Bayes, k-nearest neighbor, support vector machine, Random Forest, Extreme gradient boosting, and Neural net) were evaluated for ternary and binary classification of skin irritation potency to develop the optimal machine learning-based prediction model. Also, the importance features for classifying skin irritation/corrosion were analyzed using the Shapley Additive exPlanations (SHAP) value to get an insight into the skin irritancy of the chemicals.

Materials and methods

Data collection

Dataset established from ChemSkin Database [15] was used for the reference chemicals which were searched from the literature over the last 20 years and reviewed for the quality of in vivo data. Additional skin irritation/corrosion data were extracted from eChemPortal (https://www.echemportal.org/echemportal/ghs-search), curated and integrated. The Chemskin DB was composed of 100 substances [15], and we compiled 748 additional data from the eChemPortal site with no overlapped duplicates. The eChemPortal dataset was further investigated with ECHA reliability score to evaluate whether in vivo skin irritation/corrosion results are reliable to be used in prediction model.

For each chemical, physicochemical descriptors were collected from ChemTunes·ToxGPS site. A total of 34 features were collected: Atoms, Bonds, BondsRot, HAcc, HAccO, HAccN, HDon, HDonO, HDonN, Ro5Viol, Ro5ViolExt, Stereo, Weight, Complex, ComplexRing, ASA, McGowan, TPSA, Dipole, Polariz, LogS, XlogP, Aspheric, Eccentric, Diameter, InertiaX, InertiaY, InertiaZ, Rgyr, Span, HoF. AM1.Cor3D.ori1, Homo.AM1.Cor3D.ori1, Lumo.AM1.Cor3D.ori1, and HomoLumoGap.AM1.Cor3D.ori1. Explanation of molecular descriptors used in modeling is in Supplementary Table 1.

The reference chemicals were classified into four groups: NC, Cat 3, Cat 2, and Cat 1 according to the UN GHS classification of the Cat 1 (Category 1) for chemical inducing skin corrosion, Cat 2 (Category 2) for skin irritation, Cat 3 (Category 3) for mild irritation and NC (No Category) for no irritation. Cat 1 can be further subdivided into Cat 1A, 1B, and 1C but ignored in this research. Also, the optional Cat 3 was regarded as NC according to the OECD TG 431.

Data processing and feature selection

The compiled data was cleaned by eliminating inappropriate values and applying strict reliability standards for data selection. First, mixtures and non-liquid substances such as solids or gases were removed. Then, substances with missing values of physicochemical information were deleted, and inconsistent data were excluded from the analysis. Pearson correlation test was conducted on 34 physicochemical features to identify strongly correlated variables (coefficient ≥ 0.9 or ≤ -0.9). One representative feature of

correlated variables was selected based on feature importance analysis and relationship to skin irritation to avoid the problem of multicollinearity. The data preprocessing process was performed by the R program. General workflow of model development is provided in Fig. 1.

Machine learning analysis

The dataset was split into a model building dataset and a model testing dataset (80:20). The homogeneity of the feature distribution between datasets was evaluated by Mann–Whitney test.

Seven machine learning algorithms were evaluated for classifying skin irritation/corrosion: logistic regression (LR), Naïve Bayes (NB), k-nearest neighbor (KNN), support vector machine (SVM), Random Forest (RF), extreme gradient boosting (XGB), and neural net (NN). The description of the seven models is in detail below (1–7).

1. Logistic regression (LR) is an extension of linear regression, which is used to estimate the association of one or more independent variables with a binary or multiclass dependent variable (categorical variable). Also, LR can estimate the probability of a particular outcome with the given value of independent variables [16, 17].
2. Naïve Bayes (NB) is a simple algorithm that uses Bayes' theorem with a strong assumption that the variables are conditionally independent given the class. In statistics, Bayes' theorem indicates the relation between the conditional and marginal probabilities of events [18].
3. k-Nearest neighbor (KNN) is a widely used algorithm in pattern recognition, which is designed to determine a class by a majority vote of its neighbors. A new data is predicted by the weighed mean of the outputs from the historical database for all the k training points [19].
4. Support vector machine (SVM) is a supervised learning algorithm, widely applied in data analyzing for recognizing subtle patterns in complex datasets. The SVM develops a decision boundary known as the hyperplane. It is widely used techniques to classify mutations specific to cancer in cancer genomic studies [20, 21]
5. Random Forest (RF) is an ensemble of learning methods that consists of multiple decision trees and each decision tree is randomly built and trained so that all the decision trees have a different experience of the problem. After all the decision trees are built, RF makes final prediction in accordance with the classification problem by voting on all the decision trees [22].
6. Extreme gradient boosting (XGB) is a decision tree-based ensemble machine learning technique, minimizing the residuals of the models, and increasing the predictivity by combining weak learners. XGB is widely used by data scientists for providing state-of-the-art results on many problems [23, 24].
7. Neural net (NN) is a biologically inspired computational model designed to mimic brain function that a neuron receives multiple signals through the synapses and sends a single signal of action potentials out through its axon. The concept of a complex pattern of inputs into a single decision is suggested to the NN method, which reduces

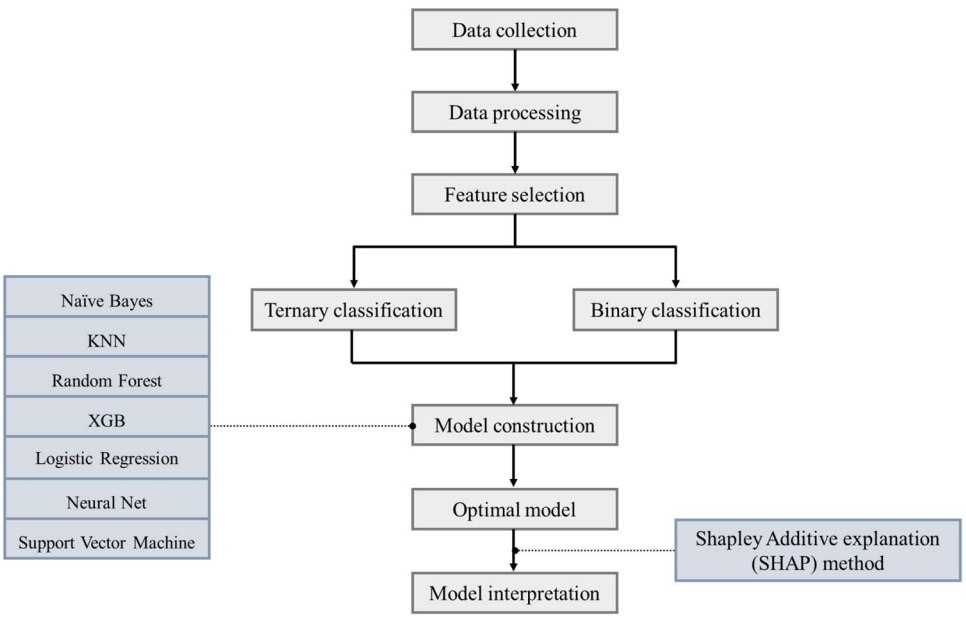


Fig. 1 General workflow of model development

complexity by comparing the inputs with the pattern of weights [25].

Each algorithm was applied to the ternary classification [Cat 1 vs. Cat 2 vs. nonirritant (NC or Cat 3)] or two binary classifications [nonirritant vs. others or Cat 1 vs. others]. The model building dataset was used for model training and hyperparameter tuning. Grid search and tenfold cross-validation were used for hyperparameter tuning, and the hyperparameter which showed high sensitivity and specificity was selected. Range of values for each hyperparameter used for the grid search was shown in Supplementary Table 2. The machine learning analysis was conducted in the Google Colab environment (https://colab.research.google.com/).

Statistical evaluation of models

The predictive performance of models was evaluated using accuracy (ACC), sensitivity (Se), specificity (Sp), positive predicted values (PPV), negative predicted values (NPV), and F_1 score. For the ternary classification algorithm, along with the overall accuracy, true and predicted classes were rearranged into two-by-two table and the performance of the binary classification (nonirritant vs others or Cat 1 vs. others) was evaluated. The statistical evaluation values were calculated by the following formula.

$$ACC = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Se = \frac{TP}{TP + FN}$$

$$Sp = \frac{TN}{TN + FP}$$

$$PPV = \frac{TP}{TP + FP}$$

$$NPV = \frac{TN}{TN + FN}$$

$$F_1 = 2 \times \frac{Se \times PPV}{Se + PPV}$$

Here, TP and TN represent true positives and negatives; FP and FN represent false positive and negatives. Se stands for the sensitivity of models, evaluating the correct prediction of positive samples. Sp is for the specificity of models, identifying the negative samples correctly. PPV means positive predicted value, and NPV means negative predicted value. These values indicate the probability of certainty of a positive or negative prediction. The F_1 score is the harmonic average of sensitivity and positive predicted value, evaluating the ability of model within the data [26, 27].

Feature importance analysis

Furthermore, the importance of features classifying skin irritation/corrosion categories were quantitatively analyzed using the SHapley Additive exPlanations (SHAP) value. Mathematically, the SHAP value corresponds to the logarithm of the probability of irritant substance relative across substances in this study. The SHAP value was calculated using SHAP package in Python program.

Results

Data characteristics

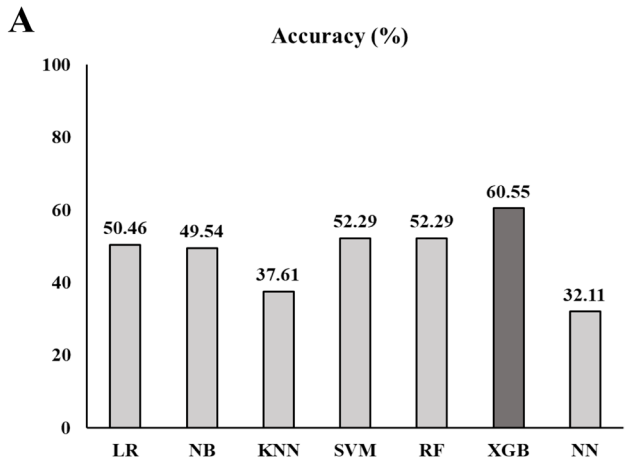
A total of 545 substances were included in this study (436 for model building dataset and 109 for test dataset). The list of all 545 liquid chemicals is provided in Supplementary Table 3. After a correlation test of 34 physicochemical features, 22 features were selected to build prediction models: HAccO, HAccN, HDon, HDonO, HDonN, Ro5Viol, Stereo, Complex, ComplexRing, TPSA, Dipole, Polariz, LogS, Aspheric, Eccentric, InertiaX, InertiaY, Rgyr, HoF.AM1.Cor3D.ori1, Homo.AM1.Cor3D.ori1, Lumo.AM1.Cor3D.ori1, and HomoLumoGap.AM1.Cor3D.ori1. There was no difference in the distribution of features between the model building and test datasets (Supplementary Table 4).

Ternary models (Cat 1 vs. Cat 2 vs. Cat 3/No Cat)

In this study, we built seven ternary models for skin irritation/corrosion and compared the model performances. Figure 2a shows the accuracy of each machine learning algorithm, and the XGB model demonstrated the highest accuracy of 60.55% for the prediction of skin irritant. The performance metrics for XGB model was presented in Fig. 2b, and showed a favorable performance.

Binary models (irritant vs. nonirritant, and Cat 1 vs. others)

Using the data, binary models for skin irritation [nonirritant (NC/Cat 3) vs others (Cat 1/Cat 2)] and binary models for skin corrosion [Cat 1 vs others (Cat 2/Cat 3/NC)] were generated with seven machine learning algorithms. Figure 3 shows two binary classification results. The XGB model showed higher accuracy than other models for both



B

True Class

Predicted Class	Cat 1	Cat 2	Cat 3 / NC
Cat 1	23	10	2
Cat 2	9	30	16
Cat 3 / NC	0	6	13

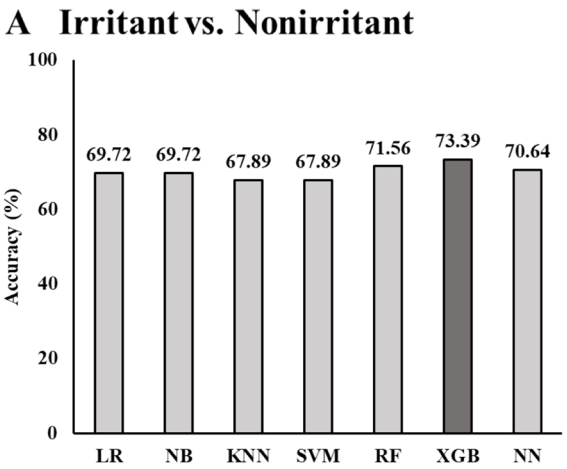
Fig. 2 Comparison of the ternary model performance of seven machine learning algorithms. **a** Accuracy, **b** confusion matrix of the XGB model

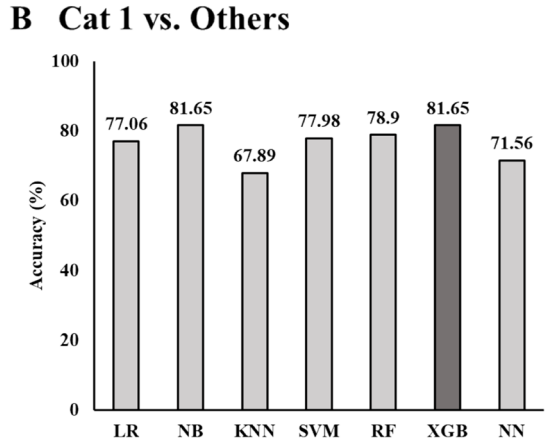
binary classifications (73.39% for irritant vs. nonirritant and 81.65% for Cat 1 vs. others). The performance metrics for XGB model was presented in Fig. 3c, d.

Performance of best working models, XGB

In the model comparison step, models generated using the XGB algorithm presented higher accuracy than other six algorithms. With the ternary and binary XGB models, statistical evaluation analysis was conducted to further investigate the performance of XGB for prediction of skin irritation or corrosion. The ternary model was rearranged into two-by-two table like the performance of the binary classification (irritant vs. nonirritant and Cat 1 vs. others).

Table 1 shows that the accuracy values of XGB models ranged from 0.7339 to 0.8165 which is comparable to the TG 439 or TG 431 which shows 0.75–0.79. Also, the sensitivity value range was 0.7188–0.9231 and the specificity value range was 0.4194–0.8571. All models for skin





C

True Class

Predicted Class	Irritant	Non irritant
Irritant	64	15
Non irritant	14	16

D

True Class

Predicted Class	Cat 1	Others
Cat 1	23	11
Others	9	66

Fig. 3 Comparison of the ternary model performance of seven machine learning algorithms. **a** Accuracy of first binary classification, **b** accuracy of second binary classification, **c**, **d** shows each confusion matrix of the XGB model

Table 1 Statistical characteristics of ternary and binary XGB models for skin irritation/corrosion

Class	ACC	Se	PPV	Sp	NPV	F_1
Ternary model performance generated with XGB algorithm						
Irritant vs. nonirritant	0.7798	0.9231	0.80	0.4194	0.6842	0.8572
Cat 1 vs. others	0.8073	0.7188	0.6571	0.8442	0.8784	0.6866
Binary model performance generated with XGB algorithm						
Irritant vs. nonirritant	0.7339	0.8205	0.8101	0.5161	0.5333	0.8153
Cat 1 vs. others	0.8165	0.7188	0.6764	0.8571	0.88	0.6970

Performance of in vitro test methods

Class	ACC	Se	Sp	Method
Irritant vs. nonirritant	0.75	0.8	0.7	TG439
Cat 1 vs. others	0.79	0.85	0.7	TG431

irritation/corrosion showed higher sensitivity values than specificity. Likewise, the PPV values ranged from 0.6571 to 0.8101 and the NPV values ranged from 0.5333 to 0.88. The overall PPV values are higher than 0.60 indicating the ability of models to accurately predict toxic compounds would be correct with more than 60% certainty.

Feature importance

Figure 4a shows the SHAP summary plot for the full UN GHS category of skin irritation/corrosion classification. Figure 4b demonstrates the SHAP summary plot for the classification of irritant and nonirritant. Homo.AM1.Cor3D.ori1, HAccN, Lumo.AM1.Cor3D.ori1, and HomoLumoGap.AM1.Cor3D.ori1 had the largest influence on classifying irritant and nonirritant substances. The SHAP values according to the change in the value of each feature are presented in the SHAP dependence plots (Fig. 5). A substance with a Homo.AM1.Cor3D.ori1 value around -10 or greater than -9 had a negative SHAP value and was therefore likely to be nonirritant and Homo value smaller than -11 was likely to be irritant. However, a substance with HOMO values between − 11 and − 9 (excluding − 10) was both likely to be irritant and nonirritant. In the case of HAccN, it was highly likely that 0 was nonirritant and 1 or more was irritant. When the Lumo.AM1.Cor3D.ori1 value was between 1 and 2, the substance was likely to be nonirritant and the Lumo value smaller than 0 was likely to be irritant. In addition, when the HomoLumoGap.AM1.Cor3D.ori1 value was -10 or more, the substance was likely to be nonirritant and the HomoLumoGap value smaller than − 11.5 was likely to be irritant.

Figure 4c shows the SHAP summary plot for classification of Cat 1 and others. Important features for classification were HAccN, Homo.AM1.Cor3D.ori1, TPSA, and Lumo.AM1.Cor3D.ori1. The SHAP values according to the change in the value of each feature are presented in the SHAP dependence plots (Fig. 6). Substances most likely to be Cat 1 were HAccN of 1 or more, Homo.AM1.Cor3D.ori1 value less than − 11, TPSA between 10 and 60, and Lumo.AM1.Cor3D.ori1 less than 0.

Discussion

This study aimed to develop computational models for predicting the GHS categories of skin irritation/corrosion of liquid chemicals with physicochemical descriptors. To ensure the high quality of training/test dataset, data was thoroughly processed and curated with reliable in vivo test results. Seven machine learning algorithms were applied to develop ternary and binary classification models for skin irritation/corrosion categories. XGB model showed the best performance with the accuracy of 60.6% for overall ternary classification and 73.4% or 81.7% for the binary classification of skin irritation or skin corrosion, respectively. The feature importance analysis was conducted using the SHAP value plots, which revealed the importance of Homo.AM1.Cor3D.ori1, HAccN, Lumo.AM1.Cor3D.ori1, and HomoLumoGap.AM1.Cor3D.ori1 for skin irritants and HAccN, Homo.AM1.Cor3D.ori1, TPSA, and Lumo.AM1.Cor3D.ori1 for skin corrosives.

Among the machine learning algorithms used in practice, XGB model has shown good performances on the classification tasks [28, 29]. In the ternary classification, the XGB model only demonstrated the accuracy over 60%. In the two binary classifications, the XGB models showed the accuracy of 73.39% and 81.65%, the highest predictive performance. According to the study from Chen, Tianqi, and Carlos Guestrin [24], the XGB was designed to be highly efficient, flexible, and portable since the system runs more than ten times faster than existing systems on a single machine. The innovations of XGB model includes a novel tree learning algorithms dealing with sparse data

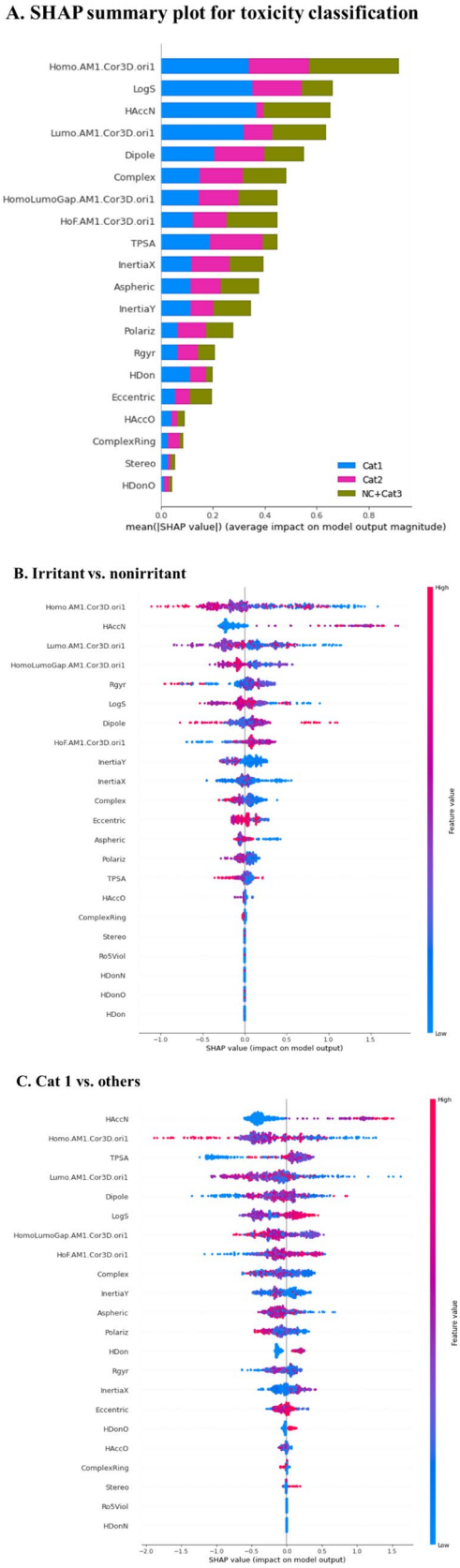


Fig. 4 SHAP summary plot for skin irritation/corrosion classification. Features in the summary plot (y-axis) are ordered by the mean absolute SHAP values (x-axis), which represents the importance of the feature in classifying GHS categories. **a** SHAP summary plot for ternary classification. The color code means the contribution of the feature for each class, **b** SHAP summary plot for binary classification (irritant vs. nonirritant). Positive SHAP values indicate higher possibilities of irritants (Cat 1 and Cat 2). Values of the feature for each substance are colored by their relative value, with red color indicating high and blue, low value, **c** SHAP summary plot for binary classification (Cat 1 vs. others). Positive SHAP values indicate higher possibilities of Cat 1

and a justified weighted quantile sketch procedure. This method dominates Kaggle competitions which competed for several machine learning problems in the world's largest data science community. The XGB model has been increasingly used to develop prediction models in the field of toxicology [30–32].

Compared to existing computational models (the accuracy of Derek Nexus 51%, and Case Ultra 59%), our model showed a significantly improved accuracy of 73.4% for the prediction of skin irritants. Furthermore, our model provides full UN GHS categorization of the skin hazard (Cat 1, Cat 2 or no cat) with a relatively good accuracy (60.6%), which will be helpful in prescreening the safety of chemicals. In addition, our model was comparable to existing in vitro skin irritation tests. OECD Test Guideline No. 431 [4] was developed for the binary classification of skin corrosion (corrosive vs. non-corrosive). TG 431 showed the accuracy of 64.58–78.75% for 80 reference chemicals according to the applied tissue models, which is comparable to our model (80.73–81.65%).

In the case of determining irritant vs. nonirritant, our ternary model showed the accuracy of 77.98% and the binary model had the accuracy of 73.39%. In the case of classifying Cat 1 vs. others, the ternary model showed the accuracy of 80.73% and the binary model showed the accuracy of 81.65% suggesting that the overall accuracy of the XGB models is comparable to the in vitro methods. Especially, the overall predictive performance of distinguishing Cat 1 (skin corrosion) had better statistical characteristics than those of TG431 or PMvar2.

McNamee, Pauline, et al. [33] reviewed the current alternative approaches for the assessment of skin irritation/corrosion and set up decision tree systems for the integration into tiered testing strategies. The initial step of decision tree for skin hazard assessment was based on the physicochemical properties and (Q)SAR to answer the question whether a test substance has the potential to be corrosive. The application of OECD (Q)SAR Toolbox [34], TOPKAT [35], DEREK, Vitic database, and Decision support system and BfR [36] was outlined. The step 2 was to conduct an in vitro corrosivity test for the identification of non-corrosive chemicals, and then the last step

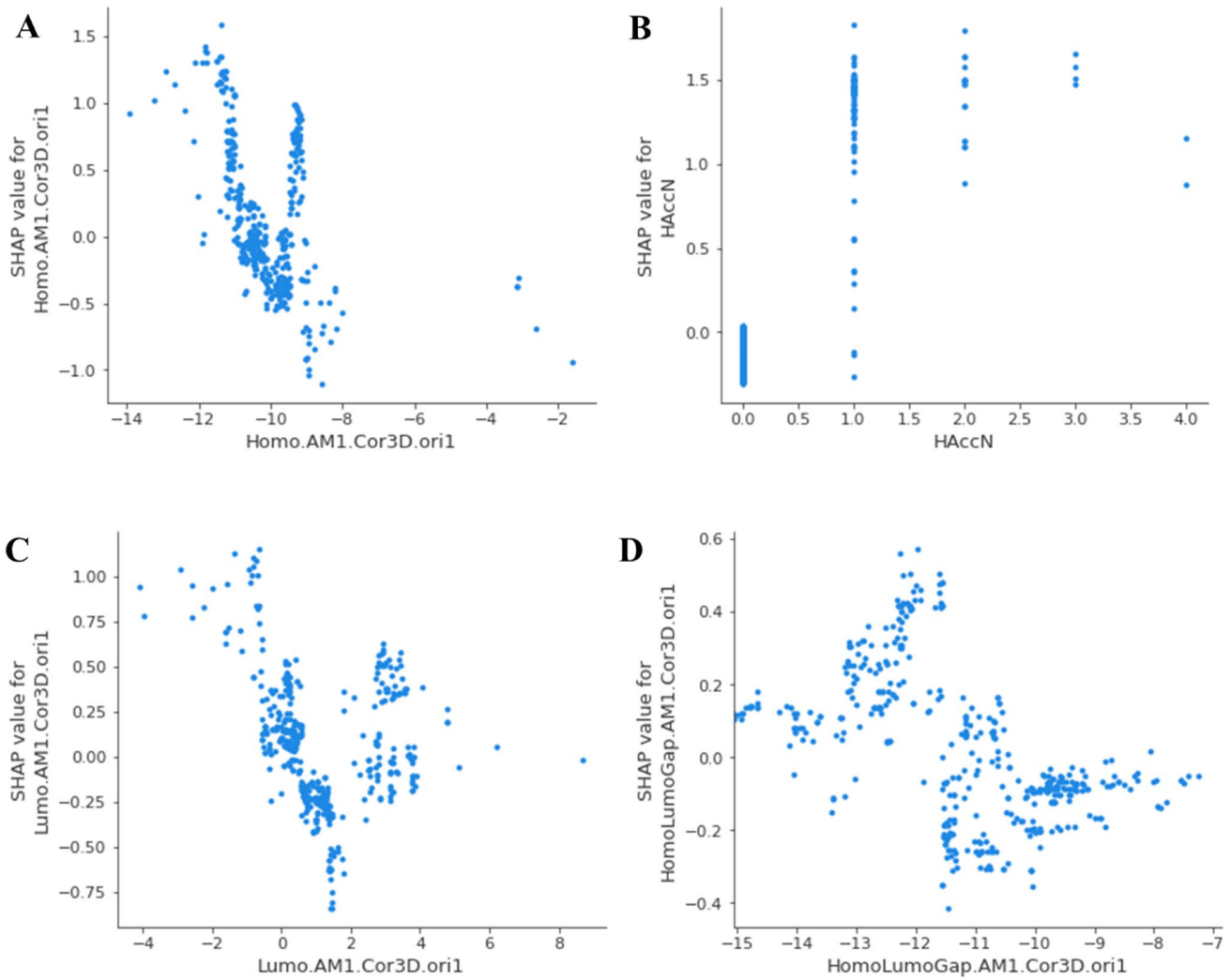


Fig. 5 The SHAP dependence plots of top 4 feature for the classification of irritant and nonirritant. **a** Homo.AM1.Cor3D.ori1, **b** HAccN, **c** Lumo.AM1.Cor3D.ori1, and **d** HomoLumoGap.AM1.Cor3D.ori1

was conducting an in vitro assay for skin irritation. Adding to the existing computational models, our model provides a holistic picture for the skin hazard of a chemical, which will be more helpful to establish confirmatory in vitro assay strategy in the frame of Integrated Approaches to Testing and Assessment (IATA).

The use of molecular descriptors in predicting the reactivity or activity of a substance has been extensively reported regarding corrosion inhibitors, drugs activity, molecular reactivity, and other response properties [37–39]. In our study, common descriptors that played an important role in distinguishing Cat 1 and irritant (or nonirritant) were HAccN, HOMO energy and LUMO energy. Toxicity has strong correlation with reactivity of chemicals [40, 41]. The presence of nitrogen atoms was a key factor for the reactive chemicals and the electronegativity had a great effect on acute toxicity [42], and increasing the molecular polarizability can strengthen chemicals' toxicity [43] supporting the importance of HAccN. Also, the chemical reactivity and kinetic stability can be explained by the distribution of the orbital frontiers. The E_{HOMO} (energy of the highest occupied molecular orbital) indicates the tendency of a molecule to donate electrons to empty molecular orbitals. The E_{LUMO} (energy of the lowest unoccupied molecular orbital) refers the ability to accept electrons. Ameh and Eddy stated that the higher value of E_{HOMO} has the better extent of reactivity or adsorption [44], supporting that HOMO energy and LUMO energy states may affect the skin hazard of chemicals.

According to the computational study of toxicity of some insecticides from Eddy and Essien [39], the correlation coefficient between quantum chemical descriptors and LD_{50} value demonstrated that the values of E_{LUMO} and HOMO–LUMO gap energy had excellent correlation with

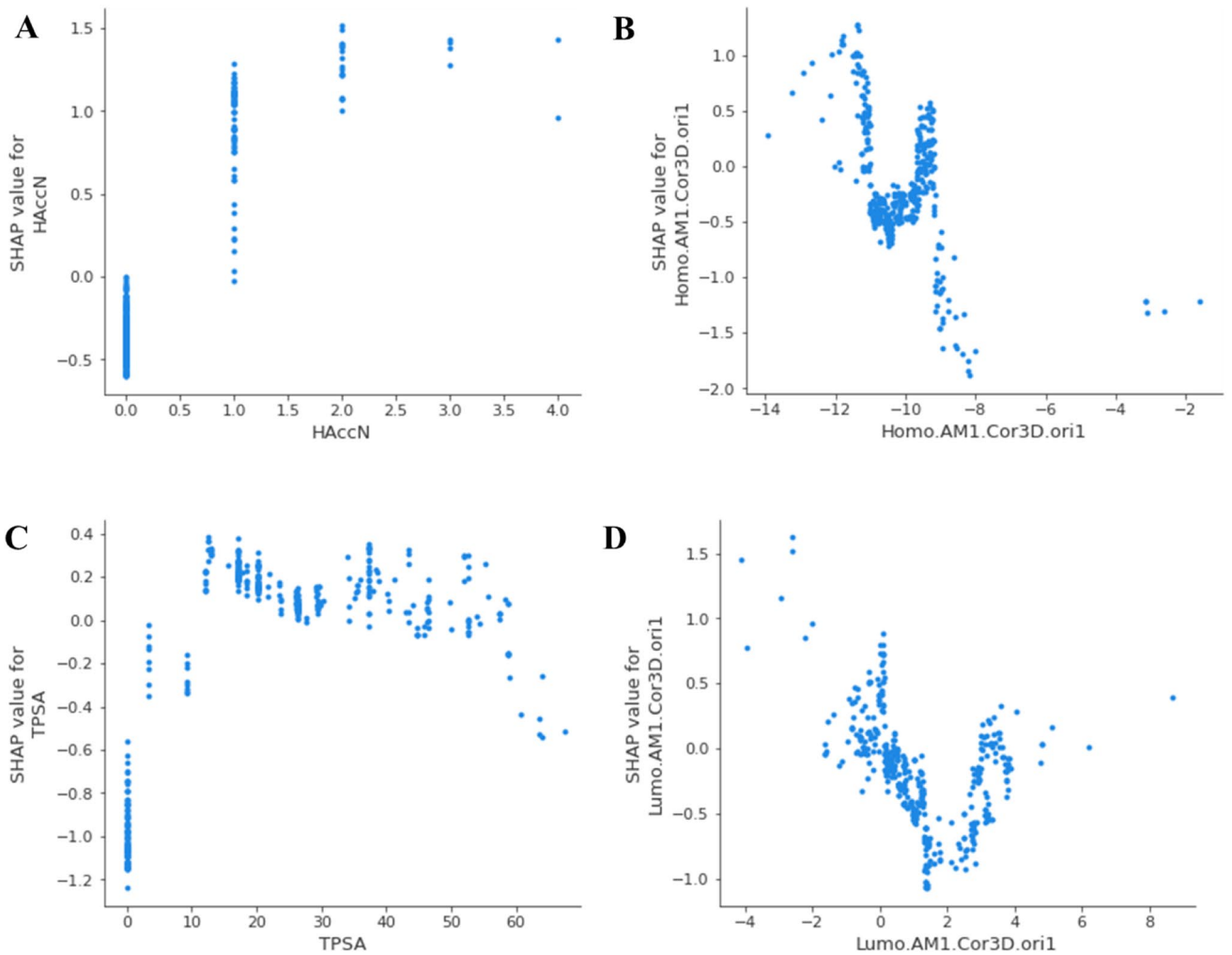


Fig. 6 The SHAP dependence plots of top 4 feature for the classification of Cat 1 and others. **a** HAccN, **b** Homo.AM1.Cor3D.ori1, **c** TPSA, and **d** Lumo.AM1.Cor3D.ori1

the toxicity values of the insecticides (respectively, 0.822 and 0.964). Also, the highest value of E_{HOMO} was obtained for CBF (which is the most toxic insecticides) having − 8.956043, while the least values were obtained for IFP (which is the least toxic) having − 9.212922. We developed the SHAP dependence plots of each descriptor using 545 liquid chemicals having wide range of feature values. E_{HOMO}, E_{LUMO}, HOMO–LUMO gap showed different tendency in each range indicating that skin irritation/corrosion cannot be simply decided by just one feature and further studies are necessary to clarify the influence of orbital energy states on the skin hazard of chemicals.

This study has several limitations. First, the number of data sufficient for machine learning analysis varies depending on the type of task, and generally the more the better. Although we collected 545 liquid chemicals, which are a large dataset, it would be supplemented with more chemicals to build a better model. Second, the predictive performance outside the range of datasets used in model development cannot be ensured. Third, our model cannot be applied to solids, gases, or diluted liquid chemicals. Solids, gases, and diluted liquid chemicals with credible skin irritation data are insufficient. Furthermore, these substances include extra steps of migration to skin or a vehicle factor in inducing skin toxicity compared to neat liquid chemicals. Accordingly, inclusion of these substances significantly decreases model performance. Notwithstanding these limitations, our model shows good performances comparable to existing in vitro assays, which could be of a great help to the assessment of skin hazards of chemicals.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s43188-022-00168-8.

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Data availability The data presented in this study are available on request to the corresponding authors.

Declarations

Conflict of interest The authors have no conflict of interest to declare.

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